METRIC
FF-B-171/10
30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, SINGLE ROW, RADIAL, NON-FILLING SLOT, CARTRIDGE TYPE WITHOUT GREASE PLUG, ANSI/AFBMA DIMENSION SERIES 33.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

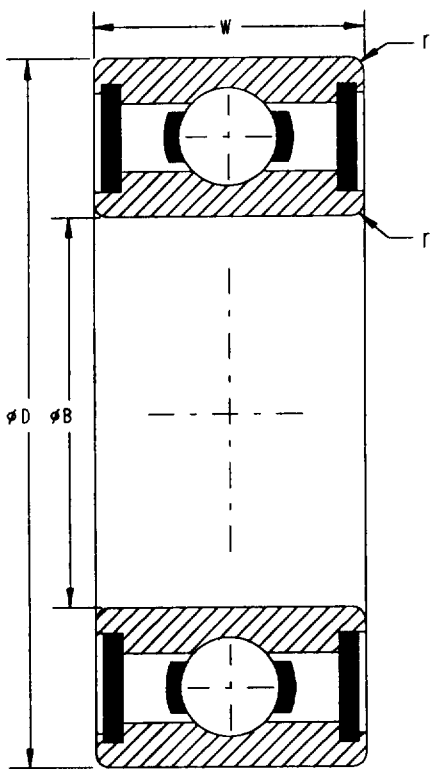


FIGURE 1.
CARTRIDGE TYPE BEARING,
DOUBLE CONTACT SEAL
(CODE A)

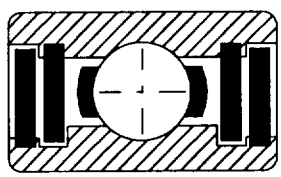


FIGURE 2.
CARTRIDGE TYPE BEARING,
DOUBLE LABYRINTH SEAL
(CODE B)

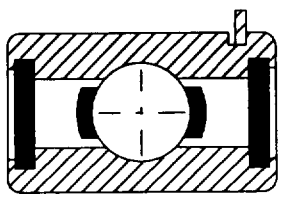


FIGURE 3.
CARTRIDGE TYPE BEARING,
DOUBLE CONTACT SEAL WITH SNAP RING
(CODE C)

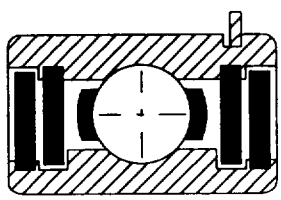


FIGURE 4.
CARTRIDGE TYPE BEARING,
DOUBLE LABYRINTH SEAL WITH SNAP RING
(CODE D)

CONFIGURATIONS OF CONTACT SEAL AND LABYRINTH SEAL VARIES AMONG MANUFACTURERS

BEARING CONFIGURATIONS

AMSC/NA FSC 3110

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REQUIREMENTS:

1. Boundary dimensions. The boundary dimensions shall be specified in the part number by the size codes in Table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Seals and snap rings. Seals shall be impervious contact type or labyrinth type and shall be specified as such. Contact seals shall be removable or nonremovable and shall be fabricated from any material unless otherwise specified. Labyrinth seals shall be removable or nonremovable and shall be fabricated from any material unless otherwise specified. Snap rings shall be fabricated from ferrous material unless otherwise specified and shall have sufficient spring temper to set firmly in the ring grooves and shall be removable without the metal taking a permanent set. The snap ring and snap ring groove dimensions and their relative locations shall be in accordance with ANSI/AFBMA Standard 20. The type of double seal and the configurations for seals and snap rings shall be specified in the part number by the codes in table III.

4. Radial Internal Clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

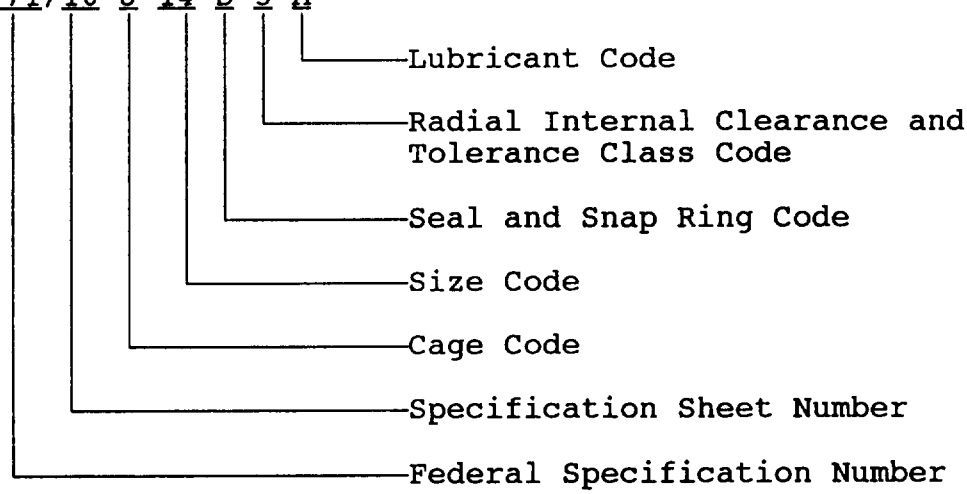
5. Tolerance Class. The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

6. Lubricant. All cartridge type bearings shall be supplied with grease. The bearing void (the airspace between the inner and outer ring of the assembled bearing) shall be 25 to 50 percent filled. The grease shall be designated by the codes "A" or "B" in the part number as specified in table V.

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7. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, seals and snap rings, radial internal clearance and tolerance class, and lubricant.

EXAMPLE: FFB171/10 J 14 D 3 A



FFB171/10J14D3A indicates - Bearing, cage-pressed steel or molded plastic; bore 70mm; outside diameter 150mm; width 63.5mm; two labyrinth seals with snap ring; radial internal clearance symbol 3 (greater than normal); tolerance class ABEC-1; lubricant in accordance with MIL-G-81322.

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic load rating (pounds)
04	20	52	22.2	1.1	27	2760
05	25	62	25.4	1.1	32	3670
06	30	72	30.2	1.1	37	4875
07	35	80	34.9	1.5	43.5	5780
08	40	90	36.5	1.5	48.5	7075
09	45	100	39.7	1.5	53.5	9160
10	50	110	44.4	2.0	60	10730
11	55	120	49.2	2.0	65	12410
12	60	130	54.0	2.1	72	14200
13	65	140	58.7	2.1	77	16000
14	70	150	63.5	2.1	82	18000
15	75	160	68.3	2.1	87	19700
16	80	170	68.3	2.1	92	22700
17	85	180	73.0	3.0	99	22900
18	90	190	73.0	3.0	104	24700
20	100	215	82.6	3.0	114	29900

1/ Size codes in Table I are industry bore codes.
2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.
3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)

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TABLE III. Codes for Seals and Snap Rings.

Code	Seal and snap ring configuration
A	double contact seal - fig.1
B	double labyrinth seal - fig.2
C	double contact seal with snap ring - fig.3
D	double labyrinth seal with snap ring - fig.4

TABLE IV. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (less than normal)	ABEC-1
2	Symbol 0 (normal)	
3	Symbol 3 (greater than normal)	
4	Symbol 4 (greater than symbol 3)	
5	Symbol 2 (less than normal)	ABEC-5
6	Symbol 0 (normal)	
7	Symbol 3 (greater than normal)	
8	Symbol 4 (greater than symbol 3)	

TABLE V. Codes for Lubricant.

Code	Lubricant
A	Grease in accordance with MIL-G-81322
B	Grease in accordance with DOD-G-24508

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force - 11

Review Activities:
Army - AR, EA, MI
Navy - OS

User Activity:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

PREPARING ACTIVITY:
DLA-IS

(Project 3110-0848-10)

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